



MANIPAL ACADEMY OF HIGHER EDUCATION

Manipal Institute of Technology, Bengaluru

BTech VI Semester

End Semester Examination - April / May -2025

SOFT COMPUTING PARADIGMS [CSE- 4054]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) A neuron receives input vector (0.8, 0.5), weights (1.0, 2.5), and bias 0.3. Use sigmoid activation to compute output. Then calculate error from target 0.9 and suggest optimization. (5)
- A)
- B) Apply the principles of hard and soft computing to a handwriting recognition system. Compare how each approach would perform in terms of accuracy, flexibility, and fault tolerance in real-world scenarios. (5)
- 2) Using K-means clustering, categorize 4 IoT devices based on power consumption and uptime. Show one iteration and centroid adjustment. (5)
- A)
- | Device | Power consumption (watts) | UpTime (Hours/Day) |
|--------|---------------------------|--------------------|
| A | 2.5 | 20 |
| B | 3.0 | 10 |
| C | 4.2 | 14 |
| D | 5.1 | 8 |
| E | 3.8 | 12 |
- B) Analyze the process of training BPNN. (5)
- 3) In a 2 x 2 SOM with input vector [0.9, 0.4] and BMU weight [0.6, 0.3], learning rate = 0.15, update the BMU's weight. (5)
- A)
- B) Compare Mamdani and Sugeno models of FIS. Discuss their mechanisms for fuzzification, rule inference, and defuzzification. (5)
- 4) (5)
- A)

A fuzzy system adjusts Air Conditioner (AC) Speed based on Room Temperature and Occupancy. The fuzzy sets are:

Room Temperature: {Cool, Warm, Hot}

Occupancy: {Few, Moderate, Full}

AC Speed: {Low, Medium, High}

Rules:

IF Room Temperature is Hot AND Occupancy is Full THEN AC Speed is High

IF Room Temperature is Warm AND Occupancy is Moderate THEN AC Speed is Medium

IF Room Temperature is Cool OR Occupancy is Few THEN AC Speed is Low

Input values:

Temperature = 82°F → $\mu_{\text{Hot}} = 0.8$, $\mu_{\text{Warm}} = 0.2$

Occupancy = 6 people → $\mu_{\text{Full}} = 0.7$, $\mu_{\text{Moderate}} = 0.3$, $\mu_{\text{Few}} = 0$

Compute the firing strength for each rule. Explain briefly how the final AC Speed decision would be influenced.

- B) You are given chromosomes of length 8 (strings are represented as abcdefgh) in a genetic algorithm. Evaluate fitness and perform one round of computation as per SGA.

Fitness function: $f(x) = (a \times b) + (c - d) + (e + f) - (g \times h)$

Chromosomes:

- $x_1 = 5\ 2\ 6\ 3\ 1\ 4\ 2\ 2$
- $x_2 = 7\ 1\ 5\ 1\ 2\ 6\ 1\ 3$
- $x_3 = 4\ 3\ 7\ 2\ 3\ 5\ 4\ 2$
- $x_4 = 8\ 2\ 3\ 1\ 4\ 2\ 0\ 1$

(5)

- 5) Analyze how Genetic Algorithms simulate evolution through selection, crossover, and mutation. Relate it to solving optimization problems.

(5)

A)

- B) Compare ANFIS and FIS with a suitable example showing input-output mapping in a system like rainfall prediction.

(5)

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